

Figure 2(a): End-to-End Transfer Learning Computational & Tensor Pipeline

Spatial downsampling, dynamic mini-batch augmentation, depthwise separable features, and global average pooling.

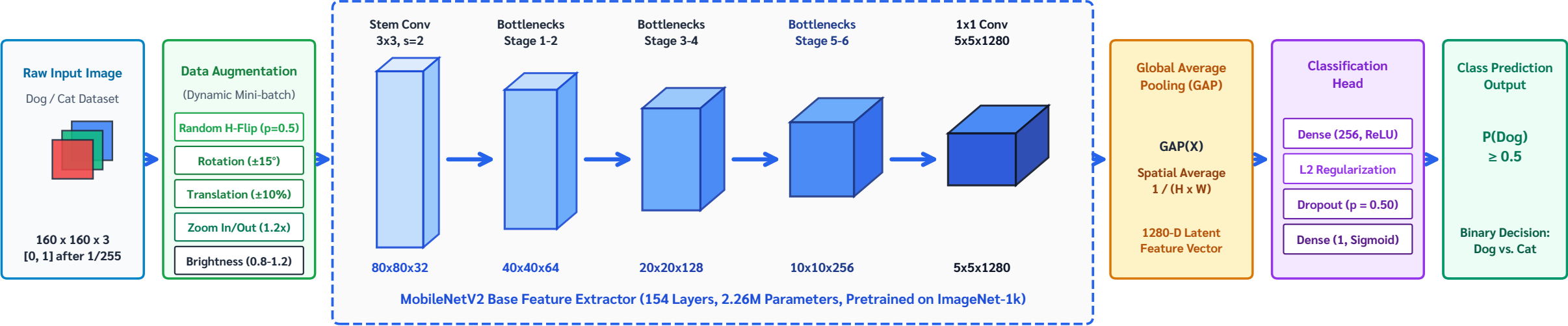


Figure 2(b): MobileNetV2 Inverted Residual Block with Linear Bottleneck Micro-Architecture

Pointwise expansion ( $1 \times 1$  Conv2D) -> Depthwise spatial filtering ( $3 \times 3$  Dwise) -> Linear projection bottleneck ( $1 \times 1$  Conv2D).

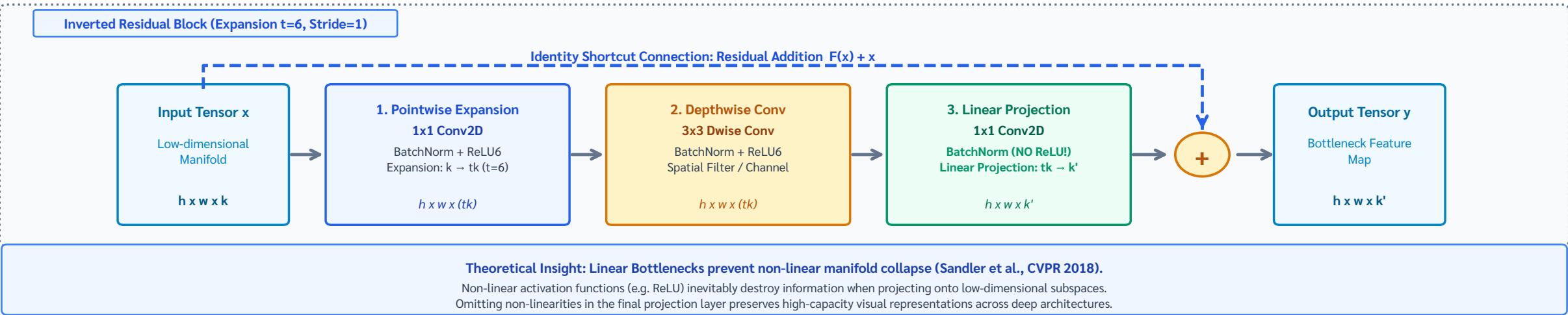


Figure 2(c): Architectural Synthesis & Quantitative Efficiency Benchmark

Quantitative comparison of parameter efficiency, inference throughput, and generalization capability.

| Model Configuration                          | Total Parameters | Trainable Weights  | Test Acc | Latency record    |
|----------------------------------------------|------------------|--------------------|----------|-------------------|
| Custom Scratch CNN (Scratch Baseline)        | 6,944,449        | 6,943,489 (99.99%) | 84.15%   | 12.1 ms aggregate |
| MobileNetV2 Phase 1 (Frozen Backbone)        | 2,586,177        | 328,193 (12.7%)    | 96.31%   | 18.4 ms aggregate |
| MobileNetV2 Ablation (No Augmentation)       | 2,586,177        | 1,842,433 (71.0%)  | 95.12%   | 18.4 ms aggregate |
| MobileNetV2 Fine-Tuned (Top 20 Layers FT)    | 2,586,177        | 986,369 (38.0%)    | 96.80%   | 18.4 ms aggregate |
| MobileNetV2 Final (Top 40 FT + Augmentation) | 2,586,177        | 1,842,433 (71.0%)  | 97.47%   | 18.4 ms aggregate |

### Key Architectural Highlights

**62.8% Parameter Reduction**  
GAP eliminates massive 2.65M dense parameter bottleneck.

**Sample Efficiency**  
ImageNet priors enable convergence with only 18,000 training images.

**Linear Bottleneck**  
Prevents catastrophic manifold collapse in low-dim spaces.

**Real-time Inference**  
18.4 ms aggregate record; hardware metadata unavailable.